

AI Career Navigator

Prompt Engineering Project

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Course: GenAI & LLMs

1. Objective

The objective is to design a focused career-advice assistant using layered prompt-engineering techniques. The project demonstrates how role instructions, response rules and few-shot examples can be combined to produce consistent guidance.

2. Prompt Design

Layer	Purpose
Role Prompt	Defines CareerNavigator as a practical career mentor.
Instruction Prompt	Controls answer structure, clarity and actionability.
Few-Shot Examples	Shows expected response patterns through examples.
Intent Router	Maps common career questions to a focused track.

3. Functional Modules

- AI / Machine Learning roadmap
- Data Analytics roadmap
- Cybersecurity roadmap
- Software Development roadmap
- Resume & Portfolio guidance
- Transition-to-IT guidance
- General career planning fallback

4. Testing

The project includes test cases for AI/ML, data analytics, cybersecurity, software development and resume queries. An unknown query is routed to general career planning guidance.

5. Conclusion

AI Career Navigator demonstrates a reproducible prompt-engineering workflow without requiring an external API key. The project separates prompt design from the local response engine, making the structure easy to extend later with a real LLM API.